Prateek Prakash Bolagond

+91-9148878006 | prateekbolagond24@gmail.com | LinkedIn

Professional Summary

Electronics and Communication Engineering student with hands-on experience building embedded systems that combine sensor integration, real-time signal processing, and on-device machine learning. Comfortable working across the stack, from firmware on ARM Cortex-M4 and ESP32 platforms to the Python/dashboard layer that sits on top. Seeking an Embedded Systems internship to apply this experience and grow further in firmware and hardware-software integration.

Education

University of Visvesvaraya College of Engineering (UVCE) 2023–2027
Bachelor of Technology in Electronics and Communication Engineering *CGPA: 8.18/10*
Bengaluru, Karnataka

Tungal Science Composite P U College 2021–2023
Pre-University Course (Class XII) *95.0%*
Jamkhandi, Karnataka

DPK Central School (CBSE) 2018–2021
10th Grade *89.02%*
Bagewadi, Karnataka

Technical Skills

- **Languages:** C, Embedded C, Python, Verilog HDL
- **Platforms:** ARM Cortex-M4 (STM32F446RE), ESP32 / ESP32-CAM
- **Interfaces:** I2C, SPI, UART, ADC (multi-channel scanning)
- **Coursework:** Analog Electronics, Digital Electronics
- **Software/Tools:** STM32CubeIDE, Arduino IDE, MATLAB, Git/GitHub

Projects

Predictive Maintenance System for Industrial Motors *STM32F446RE, Embedded C, Python*
Co-developed with a project partner 2026

- Built a low-cost real-time energy monitoring and predictive maintenance system for industrial motors, designed to catch developing faults early at a cost small manufacturers could deploy.
- Implemented non-invasive current sensing with an SCT-013 current transformer and a custom 1.65V DC-bias front end into a multi-channel-scanning ADC, computing true RMS current continuously without rewiring or downtime.
- Fused four sensor streams, current, vibration, acoustic, and temperature, into per-cycle health features using a non-blocking firmware architecture that keeps fast sensors running while slower reads complete.
- Delivered a Python bridge and live phone-accessible dashboard over local WiFi, with adaptive statistical thresholding that baselines a healthy machine and calibrates fault limits to it.

Real-Time Automated Billing System *ESP32-CAM, Embedded ML, Edge Impulse*
Co-developed with a project partner 2025

- Built a system that recognizes products as they're added to a cart and bills them in real time, removing manual scanning at checkout.
- Trained a custom object detection model on a self-built image dataset via Edge Impulse and deployed it to run fully on-device on an ESP32-CAM (AI Thinker), with no cloud dependency and low latency.
- Built the live camera capture and on-device detection pipeline running entirely on the microcontroller.
- Exposed detection results through a CORS-enabled JSON API over a WiFi web server on the ESP32, decoupling the billing display from the firmware.

Certifications

- Automateum 2.0 – Hands-on IoT Workshop, BMSCE IEEE PES & Sensors Council, November 2024
- Basics of VLSI Design, Campus Champs, 2025
- Embedded C, Udemy